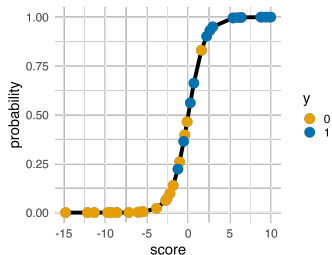
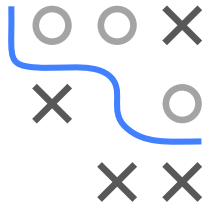


Introduction to Machine Learning

Classification

Logistic Regression



Learning goals

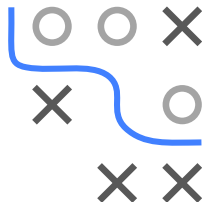
- Hypothesis space of LR
- Log-Loss derivation
- Intuition for loss
- LR as linear classifier

MOTIVATION I

- Let's build a **discriminant** approach, for binary classification, as a probabilistic classifier $\pi(\mathbf{x} \mid \boldsymbol{\theta})$
- We encode $y \in \{0, 1\}$ and use ERM:

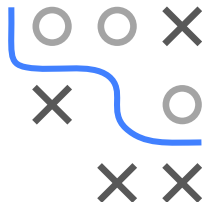
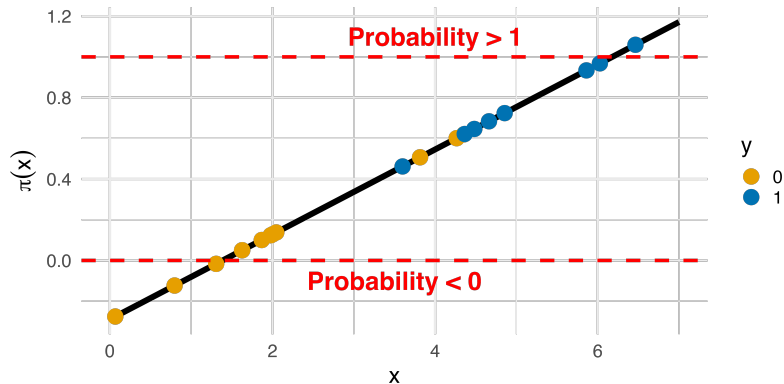
$$\arg \min_{\boldsymbol{\theta} \in \Theta} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta} \in \Theta} \sum_{i=1}^n L\left(y^{(i)}, \pi\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

- We want to “copy” over ideas from linear regression
- In the above, our model structure should be “mainly” linear and we need a loss function



DIRECT LINEAR MODEL FOR PROBABILITIES I

We could directly use an LM to model $\pi(\mathbf{x} \mid \theta) = \theta^\top \mathbf{x}$.
And use L2 loss in ERM.

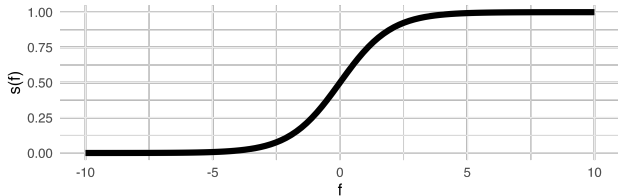


But: This obviously will result in predicted probabilities $\pi(\mathbf{x} \mid \theta) \notin [0, 1]$!

HYPOTHESIS SPACE OF LR I

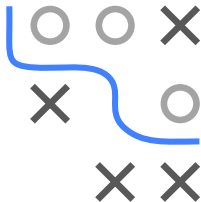
To avoid this, logistic regression “squashes” the estimated linear scores $\theta^\top \mathbf{x}$ to $[0, 1]$ through the **logistic function** s :

$$\pi(\mathbf{x} \mid \boldsymbol{\theta}) = \frac{\exp(\boldsymbol{\theta}^\top \mathbf{x})}{1 + \exp(\boldsymbol{\theta}^\top \mathbf{x})} = \frac{1}{1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x})} = s(\boldsymbol{\theta}^\top \mathbf{x}) = s(f(\mathbf{x}))$$



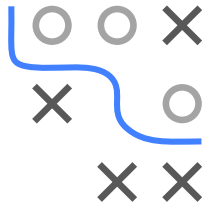
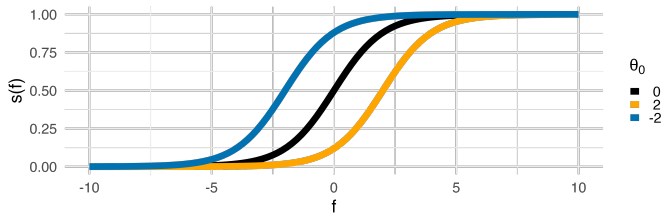
⇒ **Hypothesis space** of LR:

$$\mathcal{H} = \left\{ \pi : \mathcal{X} \rightarrow [0, 1] \mid \pi(\mathbf{x} \mid \boldsymbol{\theta}) = s(\boldsymbol{\theta}^\top \mathbf{x}) \mid \boldsymbol{\theta} \in \mathbb{R}^{p+1} \right\}$$

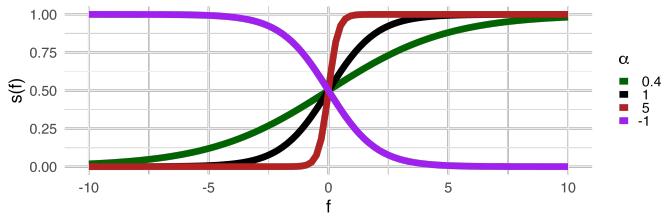


LOGISTIC FUNCTION I

Intercept θ_0 shifts $\pi = s(\theta_0 + f) = \frac{\exp(\theta_0 + f)}{1 + \exp(\theta_0 + f)}$ horizontally

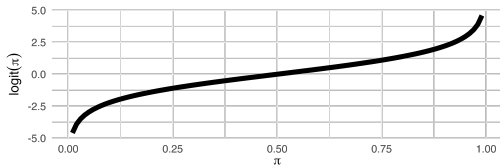


Scaling f like $s(\alpha f) = \frac{\exp(\alpha f)}{1 + \exp(\alpha f)}$ controls slope and direction



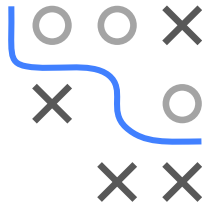
THE LOGIT I

The inverse $s^{-1}(\pi) = \log\left(\frac{\pi}{1-\pi}\right)$ where π is a probability is called **logit** (also called **log odds** since it is equal to the logarithm of the odds $\frac{\pi}{1-\pi}$)



- Positive logits indicate probabilities > 0.5 and vice versa
- E.g.: if $p = 0.75$, odds are 3 : 1 and logit is $\log(3) \approx 1.1$
- Features $\mathbf{x}$ act linearly on logits, controlled by coefficients $\boldsymbol{\theta}$:

$$s^{-1}(\pi(\mathbf{x})) = \log\left(\frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})}\right) = \boldsymbol{\theta}^T \mathbf{x}$$



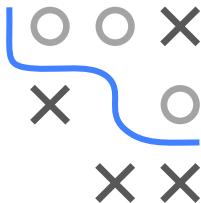
DERIVING LOG-LOSS I

We need to find a suitable loss function for **ERM**. We look at likelihood which multiplies up $\pi(\mathbf{x}^{(i)} | \boldsymbol{\theta})$ for positive examples, and $1 - \pi(\mathbf{x}^{(i)} | \boldsymbol{\theta})$ for negative.

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i: y^{(i)}=1} \pi(\mathbf{x}^{(i)} | \boldsymbol{\theta}) \prod_{i: y^{(i)}=0} (1 - \pi(\mathbf{x}^{(i)} | \boldsymbol{\theta}))$$

We can now cleverly combine the 2 cases by using exponents (note that only one of the 2 factors is not 1 and “active”):

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^n \pi(\mathbf{x}^{(i)} | \boldsymbol{\theta})^{y^{(i)}} (1 - \pi(\mathbf{x}^{(i)} | \boldsymbol{\theta}))^{1-y^{(i)}}$$



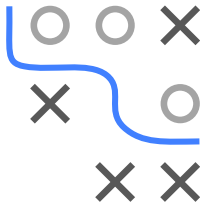
DERIVING LOG-LOSS CONTINUED I

Taking the log to convert products into sums:

$$\begin{aligned}\ell(\boldsymbol{\theta}) &= \log \mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^n \log \left(\pi \left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta} \right)^{y^{(i)}} \left(1 - \pi \left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta} \right) \right)^{1-y^{(i)}} \right) \\ &= \sum_{i=1}^n y^{(i)} \log \left(\pi \left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta} \right) \right) + \left(1 - y^{(i)} \right) \log \left(1 - \pi \left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta} \right) \right)\end{aligned}$$

Since we want to minimize the risk, we work with the negative $\ell(\boldsymbol{\theta})$:

$$-\ell(\boldsymbol{\theta}) = \sum_{i=1}^n -y^{(i)} \log \left(\pi \left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta} \right) \right) - \left(1 - y^{(i)} \right) \log \left(1 - \pi \left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta} \right) \right)$$

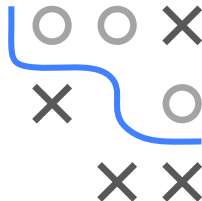
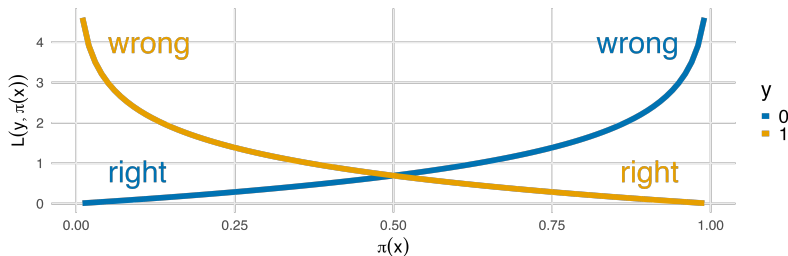


BERNOULLI / LOG LOSS I

The resulting loss

$$L(y, \pi(\mathbf{x})) = -y \log(\pi(\mathbf{x})) - (1 - y) \log(1 - \pi(\mathbf{x}))$$

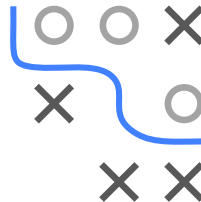
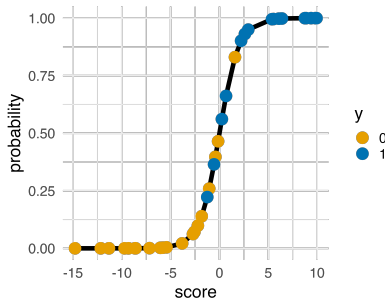
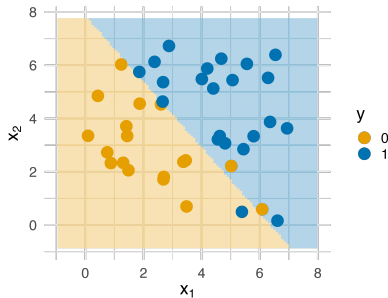
is called **Bernoulli**, **binomial**, **log** or **cross-entropy** loss



- Penalizes confidently wrong predictions heavily
- Is used for many other classifiers, e.g., in NNs or boosting

LOGISTIC REGRESSION IN 2D I

LR is a linear classifier, as $\pi(\mathbf{x} \mid \theta) = s(\theta^\top \mathbf{x})$ and s is isotonic.



OPTIMIZATION I

- Log-Loss is convex, under regularity conditions LR has a unique solution (because of its linear structure), but not an analytical one
- To fit LR we use numerical optimization, e.g., Newton-Raphson
- If data is linearly separable, the optimization problem is unbounded and we would not find a solution; way out is regularization
- Why not use least squares on $\pi(\mathbf{x}) = s(f(\mathbf{x}))$?
Answer: ERM problem is not convex anymore :(
- We can also write the ERM as

$$\arg \min_{\theta \in \Theta} \mathcal{R}_{\text{emp}}(\theta) = \arg \min_{\theta \in \Theta} \sum_{i=1}^n L(y^{(i)}, f(\mathbf{x}^{(i)} | \theta))$$

With $f(\mathbf{x} | \theta) = \theta^T \mathbf{x}$ and $L(y, f(\mathbf{x})) = -yf(\mathbf{x}) + \log(1 + \exp(f(\mathbf{x})))$

This combines the sigmoid with the loss and shows a convex loss directly on a linear function

